

FINAL EXAM

ECON 2010 – Macroeconomics

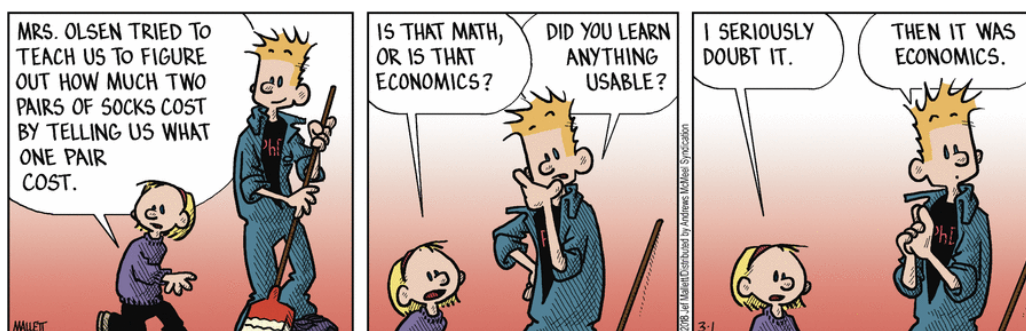
Spring 2019

To be completed in-class on Monday, May 6, 2019. You will have 2.5 hours to complete the exam (5:30-8:00pm).

Name: _____

Final exam instructions:

1. There are **two** problems on the exam, each with multiple parts.
 2. Read each problem carefully. Just because it looks like a problem you've seen before doesn't mean that the exam asks you to do exactly what you've done before.
 3. Pay particular attention to when a question asks you to explain your answer in words. Your explanations will be worth the most points of any type of question on the exam.
 4. Write your final answers as legibly as possible, and make clear what your final answer is.
 5. Show your work. Even if your answer is not exactly correct, you will receive credit for setting up the problem correctly, or describing in words how you're thinking through it. The better that I understand what you've done and why, the more likely you receive partial credit.
 6. You may not use notes, the textbook, any other materials on the exam, or confer with anyone about the exam while you're taking it. To say it differently, this is a "closed book" exam.
 7. If you think that a question is ambiguous, come up to the front of the room and ask me. If I agree that the question is ambiguous, I will clarify. If I don't agree that the question is ambiguous and you still think that there is room for interpretation, make a choice in how you interpret the question so that you can answer the question. Then *explain your interpretive choice in your answer*. I may take into account your interpretation when grading.
 8. Take a deep breath, remember how much you've learned in class, and do your best to demonstrate what you've learned!
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This exam is designed to show otherwise!

DO NOT TURN THIS PAGE UNTIL INSTRUCTED TO DO SO.

Problem 1 — Teaching Your Friend Economics (50 points)

You have a friend who thinks that they know a lot about economics. We all know someone like that, right? Unfortunately, they've never taken a class in economics. So, some of the things that they know are correct, but they're also mistaken about many other things. Now that *you've* taken an economics class, you know when what they're saying is right and when it's wrong.

Consider the following statements that your friend makes. For each statement, do two things. **First**, indicate whether the statement is **true or false**, and **second**, **briefly explain *why*** the statement is true or false.

Your explanations will be assessed on the correct logic of the explanation *why*, as well as on whether you discuss concepts (with associated terms) relevant to the statement, and discuss those concepts correctly.

Remember: You do not need knowledge from outside of class to answer these questions. The questions are designed to assess your understanding of *our course's material*. Indicating true or false correctly is worth 1 point for each statement, while the explanation is worth 4 points for each statement.

(a) *“Economics is just the study of money.”* (5 points)

(b) *“I recently read that the unemployment rate has fallen. But a fall in the unemployment rate could be good news in terms of people having jobs or actually bad news. You can’t tell whether more people have jobs just from a fall in the unemployment rate.”* (5 points)

(c) *“Yeah, I’ve heard of the U.S. Federal Reserve — they declare every month what interest rates are in the economy.” (5 points)*

(d) *“I’ve been looking for a job as a checkout clerk at a grocery store and the wages are just really terrible. The wages [in the market for grocery store checkout clerks] are determined by just whatever the grocery stores want to pay their workers, and they don’t want to pay them very much, so they don’t have to.” (5 points)*

(e) *“Someone was telling me about how India has been experiencing a lot of [nominal] GDP growth recently. But that might not be as much growth as it seems because, first, their population has been growing a lot, and, second, they’ve also had a lot of inflation.”* **(5 points)**

(f) *“I would like to study more economics because it’s a discipline that answers questions like ‘should we increase immigration’ or ‘should we abolish the Federal Reserve’, and I want to know the answers to those questions.”* **(5 points)**

(g) *“The reason that the U.S. dollar functions as money is because of the government’s gold reserves. You can always take your dollar to the government and get a fixed quantity of gold in return.”* **(5 points)**

(h) *“The commercial banking system in the U.S. doesn’t increase or decrease the amount of money in the economy, they just store people’s money.”* **(5 points)**

- (i) *“In the long run, if we want our economy to grow [i.e. produce GDP growth], we just need to have people [i.e. consumers and firms] buy more stuff [i.e. demand more output]. It’s really just that simple.” (5 points)*

- (j) *“There’s no such thing as a market for money because you can’t put a price on money. Money is money.” (5 points)*

Problem 2 – The Consequences of a Balanced Budget Amendment for Fiscal Policy and the Macroeconomy (50 points)

In recent years there have been frequent calls for a national balanced budget amendment to the U.S. Constitution. A balanced budget amendment (as typically proposed) says that the Federal government can run budget surpluses, but that the Federal government cannot run a budget deficit. If the government is projected to run a deficit, the amendment requires that the government *either* (1) increase tax revenues or (2) cut expenditures to balance the budget. Again, the government does not need to take action if it is running a surplus.

Just a year ago, an amendment to require the Federal Government to maintain a balanced budget failed in the U.S. Congress. Despite failing, the bill was supported by 233 Congressmen.

The *New York Times* quoted Robert W. Goodlatte, Republican of Virginia and supporter of the amendment, as making the following argument in favor of the amendment: “It’s time for Congress to stop saddling future generations with the burden of crushing debts to pay for current spending. We should not pass on to our children and grandchildren the bleak fiscal future that our unsustainable spending is creating.”

But is this thinking correct? Would a balanced budget amendment be a good thing for the U.S. economy? Or are there tradeoffs that politicians must also consider? In this question you’ll consider (just) some of the costs and benefits that such an amendment would impose on the macroeconomy.

DEFINITIONS:

A **budget surplus** is defined as a situation in which tax revenues exceed expenditures in a given budget year.

A **budget deficit** is defined as a situation in which expenditures exceed tax revenues in a given budget year.

A **balanced budget** is defined as a situation when tax revenues equal expenditures.

IMPORTANT: Make the following assumptions as you consider the questions in this problem:

- Assume that the tax component of fiscal policy is *only* personal income taxes (nothing related *at all* to firms/businesses in any way).
- Assume that the expenditure component of fiscal policy is *only* the salaries of government employees (i.e. paying for those employees’ productive output doing their jobs).
- Assume that if the economy is in a recession (i.e. a situation where output is *less* than potential output) that the government collects *less* tax revenues than it spends in expenditure (because more workers are unemployed, wages are lower, and so there is less personal income to tax).
- Assume that if the economy has moved into an expansionary phase (i.e. a situation where output is *above* potential output) that the government collects *more* tax revenue than ex-

penditure (because unemployment is low, wages are high, and so there is more personal income to tax).

- Assume that when output is equal to potential output tax revenue equals expenditure and so the budget is always balanced.
- As noted above, the balanced budget amendment *requires* that if the government is projected to run a deficit, it must either increase taxes or cut expenditures in order to balance the budget.
- Apply the principle of *ceteris parabis* and hold all else fixed when you are drawing your curves and considering the consequence of a given change described in the problem.

(a) Does a balanced budget amendment affect fiscal policy or monetary policy? (1 point)

(b) Consider our standard model of Aggregate Supply-Aggregate Demand (AS-AD). Consider the U.S. economy around 2008 when we were in the throes of the Great Recession. As we've discovered in a previous homework assignment, many economists believe that the Great Recession was triggered by a contraction (i.e. reduction) in Aggregate Demand (AD) brought about by — among other things — a reduction in consumer stock market wealth, a reduction in consumer housing wealth, and a decline in consumer confidence.

Given this information about the Great Recession and its causes, draw a plot of the AS-AD model that shows the U.S. economy in a **short run equilibrium** that corresponds to this interpretation of the what led to the Great Recession (i.e. this interpretation of which curve shifted and in which direction to put the economy in a recession). **Label** your short run equilibrium with an **A**. Be sure to **label the axes** of your plot appropriately and **draw all of the curves** necessary for your plot to be a plot of AS-AD as we typically do in class and in homeworks — i.e. both short and long run curves. You do *not* need to draw a long run equilibrium. To say again, you are drawing a short run equilibrium, not a long run equilibrium, but you need all short and long run curves on your plot.

(Note: This question asks you to *start* from the place that all of our in-class and homework problems *end*. If you're not sure how to draw this plot, you could get to the right place by the usual process that I walk you through in a homework assignment. Start from the economy in a long run equilibrium, figure out which curve has shifted because of a reduction in consumer wealth, figure out which way it has shifted according to the information in the problem above, draw the long run equilibrium, and then the curve shift. Finally, label your new short run equilibrium resulting from the curve shift with an A as instructed in the problem. You do not need to do all of this work, I only suggest it if you're unclear about how *start* with the plot that represents the short run equilibrium of the Great Recession.) (5 points)

(c) Is the short-run equilibrium output indicated by point **A** that you drew in part (a) **above or below** the economy's potential output? **(1 point)**

(d) Consider that we are in the short run equilibrium indicated by point **A** on your plot in part (b) above. Assume that the government **is subject to a balanced budget amendment** as described in the beginning of this problem (and that it adheres to it). That is, government expenditures may *not* exceed government tax revenue.

Given the location of point **A** in your part (b) plot, does the government need to change any aspect of its fiscal policy to be in compliance with the balanced budget amendment? **Yes or no?** If yes, **what two specific actions** can the government take in terms of adjusting fiscal policy **so that it is in compliance with** the balanced budget amendment (i.e. what can the government do to each component of fiscal policy to comply with the amendment)? **(2 points)**

- (e) Consider the AS-AD model and the short run equilibrium **A** that you drew in part (b) above. Which curve(s) in the model shift when the government makes **either** of the above changes to fiscal policy indicated in part (e) to be in compliance with the balanced budget amendment? If the government does not need to make a change, simply indicate *no curves shift*. **(2 points)**
- (f) Indicate the direction that the curve(s) you said would shift in part (e) shift, if any. For each curve, you can say *left*, *right*, or you can say *none shift*. **(2 points)**
- (g) In just a few sentences, briefly explain why the curve(s) you indicated would shift have shifted. If no curves shift, you can say *none shift*. **(5 points)**

(h) Re-draw the short-run equilibrium plot from part (b) here. Label the starting short-run equilibrium **A** (as in your plot from part (b)). Draw any curve(s) that you indicated would shift in part (e) with a dashed line. Label the *new short run* equilibrium that results from the shifted curves with a **B**. I have not given you specific information about how much the curve(s) have shifted, so simply draw a generic shift in the direction you indicated the curve would shift. If no curves shift, simply re-draw your plot from part (b) above. **(5 points)**

(i) What happens to *each* price levels and output (GDP) as a result of a shift to the equilibrium from point **A** to point **B** in your plot for part (h)? **(2 points)**

(j) Assume that the Federal government would like the economy to operate at or close to full employment (i.e. potential output). Has the equilibrium **B** produced by the balanced budget amendment helped the economy get **closer to or further from** operating at full employment relative to our starting equilibrium, point **A**? Briefly, in just a few sentences, **explain why** the balanced budget amendment has helped or hindered the economy in operating at full employment. **(4 points)**

(k) Now assume that the Federal government is **no longer subject to a balanced budget amendment**. That is, it *can* run deficits if it wants. What does the U.S. government do to finance (i.e. pay for) a deficit when one occurs? **(2 points)**

- (l) “Capital is a factor of production that determines potential output.” In just a few sentences, explain **what capital is** in economic terms, and **what that statement in quotations means**. Be sure to mention the Long Run Aggregate Supply (LRAS) curve in your answer. **(3 points)**

- (m) Explain what it means for the Marginal Propensity to Consume (MPC) in the U.S. to be 0.65. **(2 points)**

- (n) Consider the LRAS curve. Assume that when consumers save, they deposit their money in a bank that uses its liabilities to make *loans to firms*. In a few sentences, briefly explain how *consumer savings in a bank* like this can increase potential output. Another equivalent way of asking this question is: How can consumer savings shift the LRAS to the right? A further equivalent way of asking this question is: How can consumer savings contribute to economic growth?

(Note: You may want to think back to our in-class exercise with the wheat producers and the ploughs as you formulate your answer.) **(4 points)**

- (o) Now consider that consumers have a *choice* between saving their money in a bank that uses its liabilities to make loans to firms **or** by buying a government bond. Given what the beginning of the problem said about what comprises government expenditures, **and** given your explanation for how consumer savings is related to shifts in the LRAS in part (n) above, **which method of saving** (buying government bonds or saving money in a bank) is more likely to increase potential output? In just a few sentences, **briefly explain why**. **(4 points)**

(p) Now assume that the Federal government would like to generate economic growth (i.e. increase potential output, i.e. shift LRAS to the right). Consider your answers to parts (k)-(o) of this problem when the government is **not** subject to a balanced budget amendment. If the government is running a deficit, does that **help or hinder** the economy in generating long run economic growth? Briefly, in just a few sentences, **explain why** the lack of a balanced budget amendment (i.e. the fact that the government has run a deficit) has helped or hindered the economy in generating economic growth. **(4 points)**

(q) Now that you've considered some of the consequences of a balanced budget amendment, you can formulate a *normative* statement about this amendment. **Write a normative statement** about the balanced budget amendment and fiscal policy. Your statement **must include the word "because"**, followed by some sort of explanation. You aren't being judged on whether I agree or disagree with your normative statement (*you* don't even have to agree with your normative statement!); only on whether what you write successfully qualifies as normative. However, any factual (i.e. positive) components of your statement need to be accurate. **(2 points)**